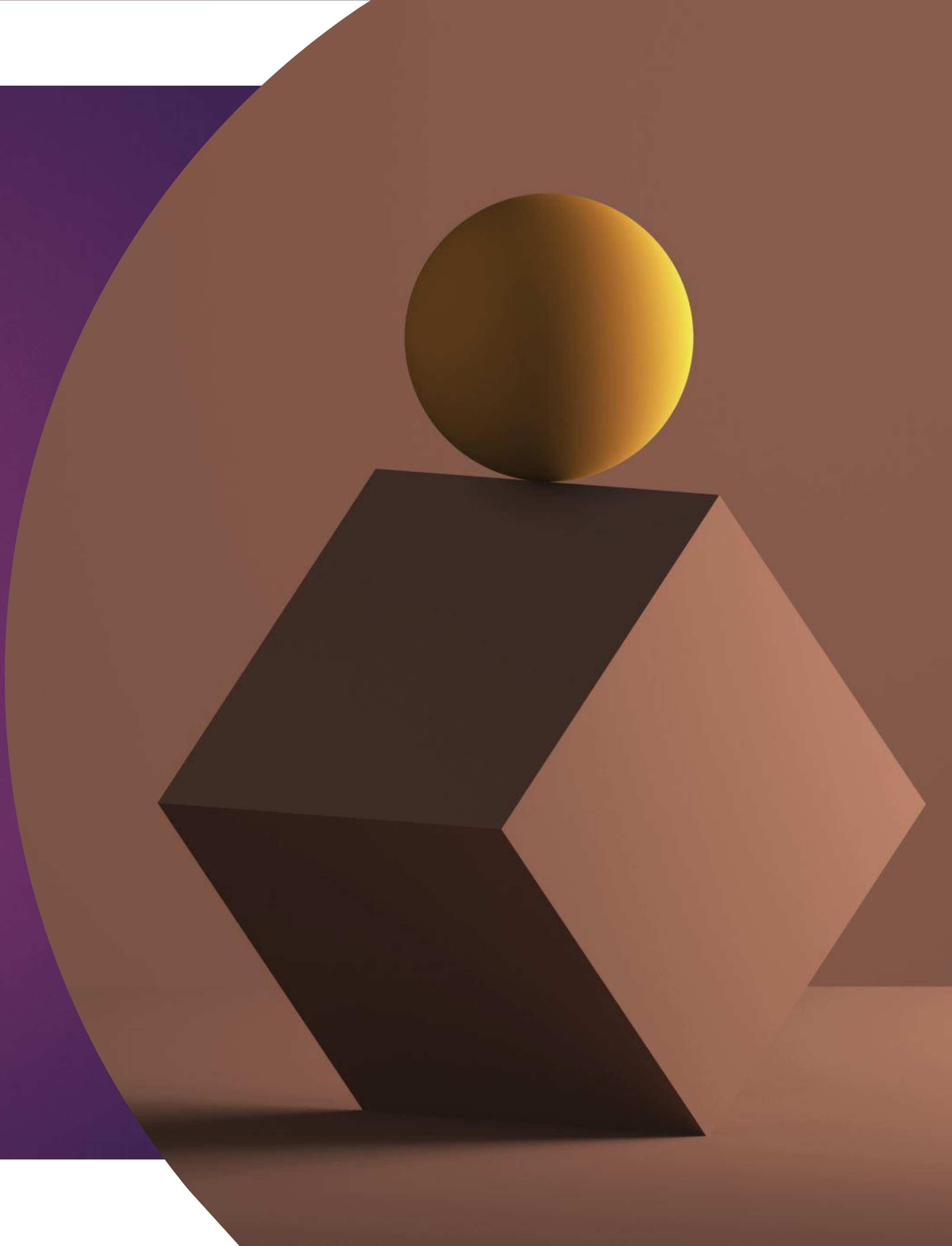
Kinetic Modelling in virtual reality



Introduction

Location	Motion	Parent child relationship
• Determines location of 3D objects with respect to world system of coordinates	• Also determines motion in the virtual world	• Governed by parent-child relationships. Motion of parent affects motion of child

Homogeneous Transformation Matrices

Homogeneous Transformation Matrices

$$\mathbf{T}_{A \leftarrow B} = \begin{bmatrix} \mathbf{R}_{3\times3} & & & \mathbf{P}_{3\times1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Generic equation

- **Denavit and Hartenberg [1955]** and later **Fu et al. [1987]** and Foley et al. [1990] chose 4x 4 homogeneous transformation matrices
- **R [3x3] is the rotation submatrix** expressing the orientation of system of coordinates B with respect to system of coordinates A
- **P [3x1]** is the vector expressing the **position** of the origin of system B with respect to the origin of system of coordinates

Homogeneous Transformation Matrices

- Systems of coordinates A and B are orthonormal, having cartesian triads of vectors (i, j, k) such that norm $||i|| = ||j|| = ||k|| = 1$
- Dot product of any two is zero

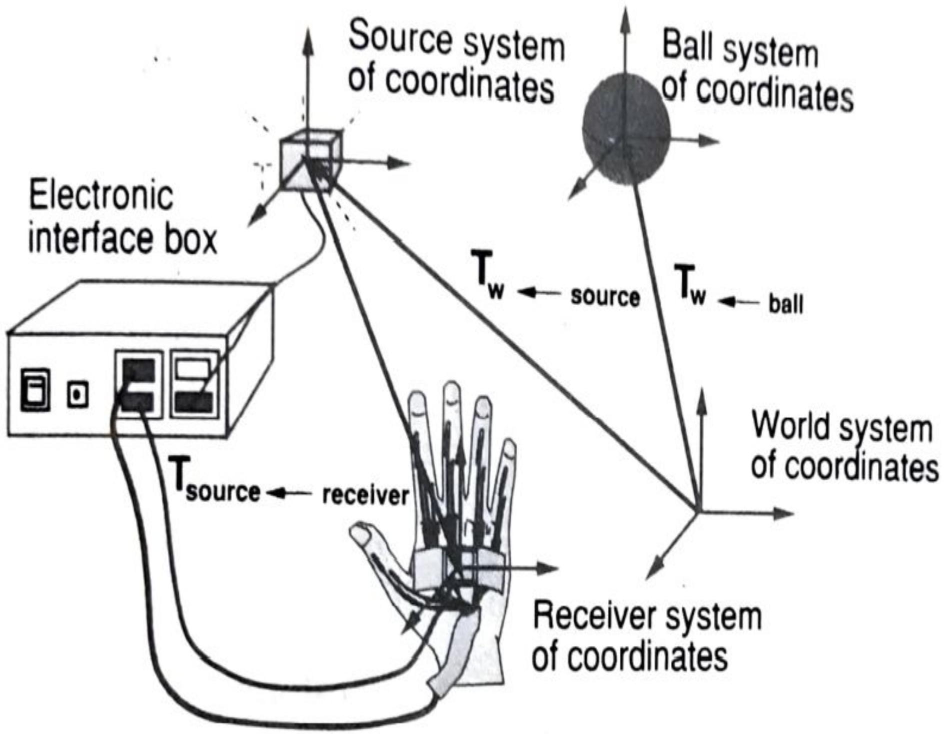
- Advantages
  - treat both rotations and translations in a **uniform way**
  - they can be **compounded**, such that complex kinematics relationships can be modeled

Object Position

Transformation Invariants

- 3D hand's position and orientation are mapped to the position and orientation of the 3D tracker attached to the sensing glove
- The **tracker is positioned with respect to a source**
- The world coordinates are positioned and scaled with respect to the source. This can be represented by the matrix $\boldsymbol{T}_{\boldsymbol{w} \leftarrow \boldsymbol{source}}$ which is time dependent
- Tracker's receiver position is represented as $\boldsymbol{T}_{\boldsymbol{source} \leftarrow \boldsymbol{receiver}}$ which is time dependent.
- Virtual hand can be moved in the virtual world by multiplying all these matrices
- $V_i^{w}(t) = \boldsymbol{T}_{\boldsymbol{w} \leftarrow \boldsymbol{source}} \boldsymbol{T}_{\boldsymbol{source} \leftarrow \boldsymbol{receiver}}$ (t) V_i^{hand}
- V_i^{hand} are the vertex coordinates in the hand system of coordinates, and i = 1…..n

sensing glove and virtual ball



$T_{w \leftarrow ball}$ - virtual ball 3D position

$T_{receiver \leftarrow ball}(t) = T_{receiver \leftarrow w}(t)\, T_{w \leftarrow ball}(t)$

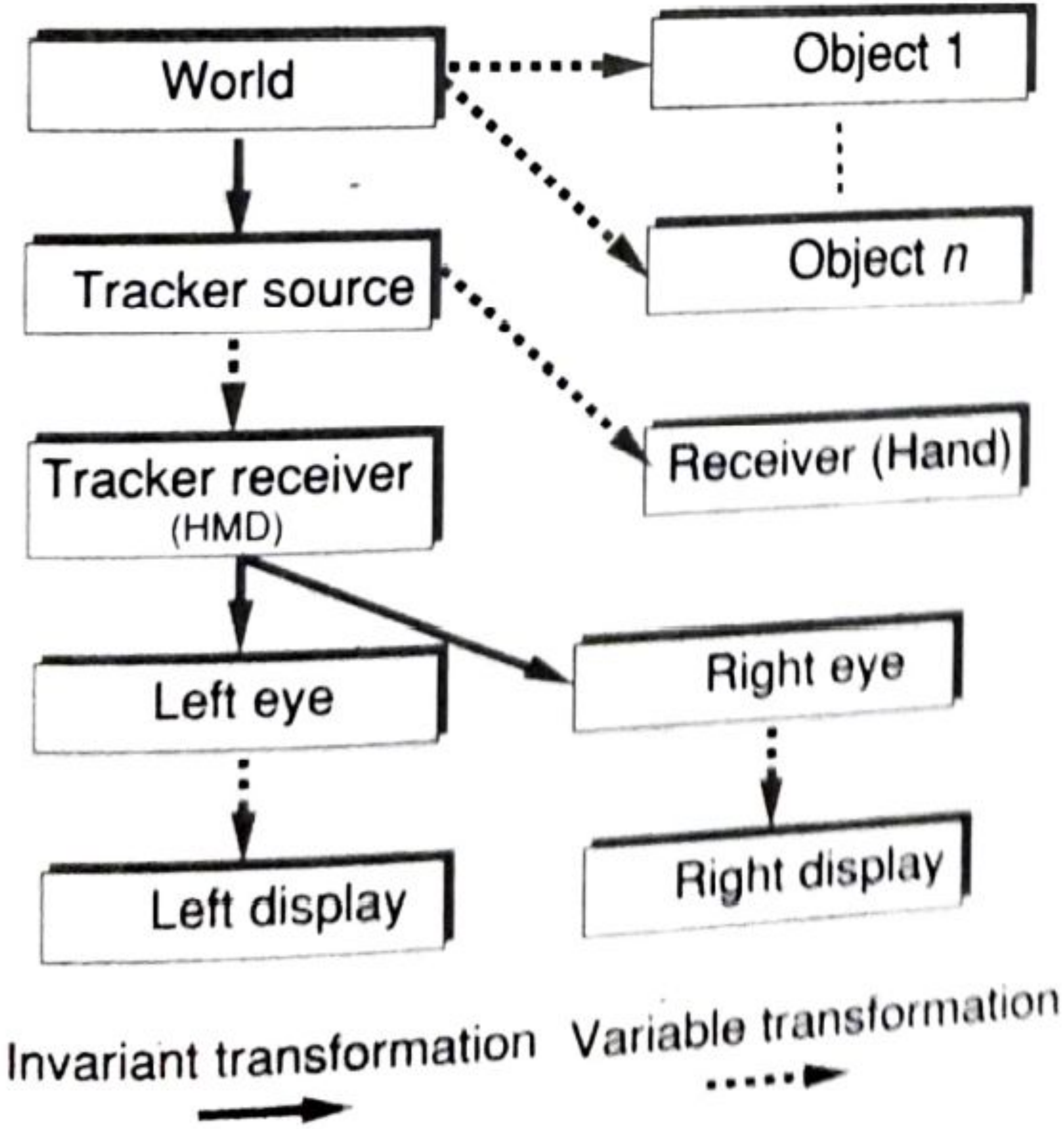
$T_{receiver \leftarrow ball}(t) = (T_{w \leftarrow receiver}(t))^{-1}\, T_{w \leftarrow ball}(t)$

Once the ball is grasped, its position does not change with respect to the receiver (Invariant)

Invariant here means that the transformation matrix is not a function of time

$$\mathbf{T}_{w \leftarrow ball}(t) = \mathbf{T}_{w \leftarrow source}\mathbf{T}_{source \leftarrow receiver}(t)\mathbf{T}_{receiver \leftarrow ball}$$

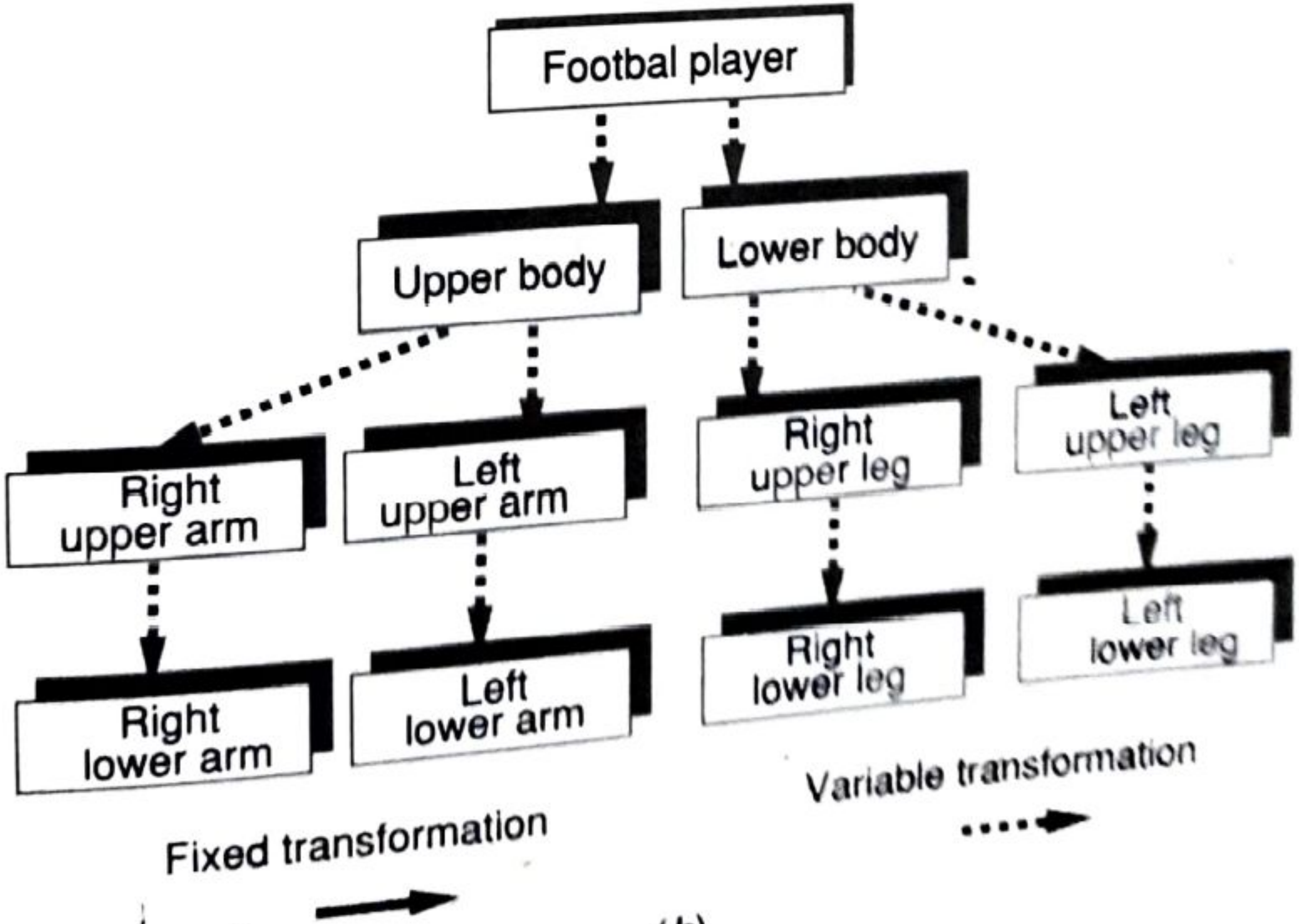
Viewing the world with an HMD



Object Hierarchies

Object Hierarchies

- **Object hierarchies define groups of objects which move together as a whole, but whose parts can also move independently**.
- The **higher-level** objects are called the **parent objects**
- **Lower-level** objects are called **child objects**
- The motion of parent object is replicated by that of all its children.
- **Child object can move independently** without affecting the parent object's motion.
- The segmented model hierarchy can be **represented by a tree graph**
- Its nodes are object segments, and its **branches represent relationships**.
- Geometric transforms applied to a given parent object are automatically passed on to all its children
- At the top of the tree graph stands a **global transformation** that determines the view to the scene. If this matrix is changed, all objects in the virtual world will appear translated, rotated, or scaled


Footbal player
Upper body
Lower body
Right upper arm
Left upper arm
Right upper leg
Left upper leg
Right lower arm
Left lower arm
Right lower leg
Left lower leg
Fixed transformation
Variable transformation

(b)

Virtual Hand example

- Its hierarchy has a palm parent and five finger children. Thus, when the palm moves (according to the tracker data). so do its children (fingers)
- The finger becomes parent to the first finger link, which becomes parent to the interphalangeal link, which in turn is parent to the last (distal) link

$$\mathbf{T}_{\mathrm{w}\leftarrow\mathrm{fingertip}}(t) = \mathbf{T}_{\mathrm{w}\leftarrow\mathrm{source}}\,\mathbf{T}_{\mathrm{source}\leftarrow\mathrm{receiver}}(t)\,\mathbf{T}_{\mathrm{receiver}\leftarrow 1}(t)\,\mathbf{T}_{1\leftarrow 2}(t)\,\mathbf{T}_{2\leftarrow 3}(t)\,\mathbf{T}_{3\leftarrow\mathrm{fingertip}}$$

$$\mathbf{T}_{\mathrm{global}\leftarrow\mathrm{fingertip}}(t) = \mathbf{T}_{\mathrm{global}\leftarrow\mathrm{w}}(t)\mathbf{T}_{\mathrm{w}\leftarrow\mathrm{fingertip}}(t)$$

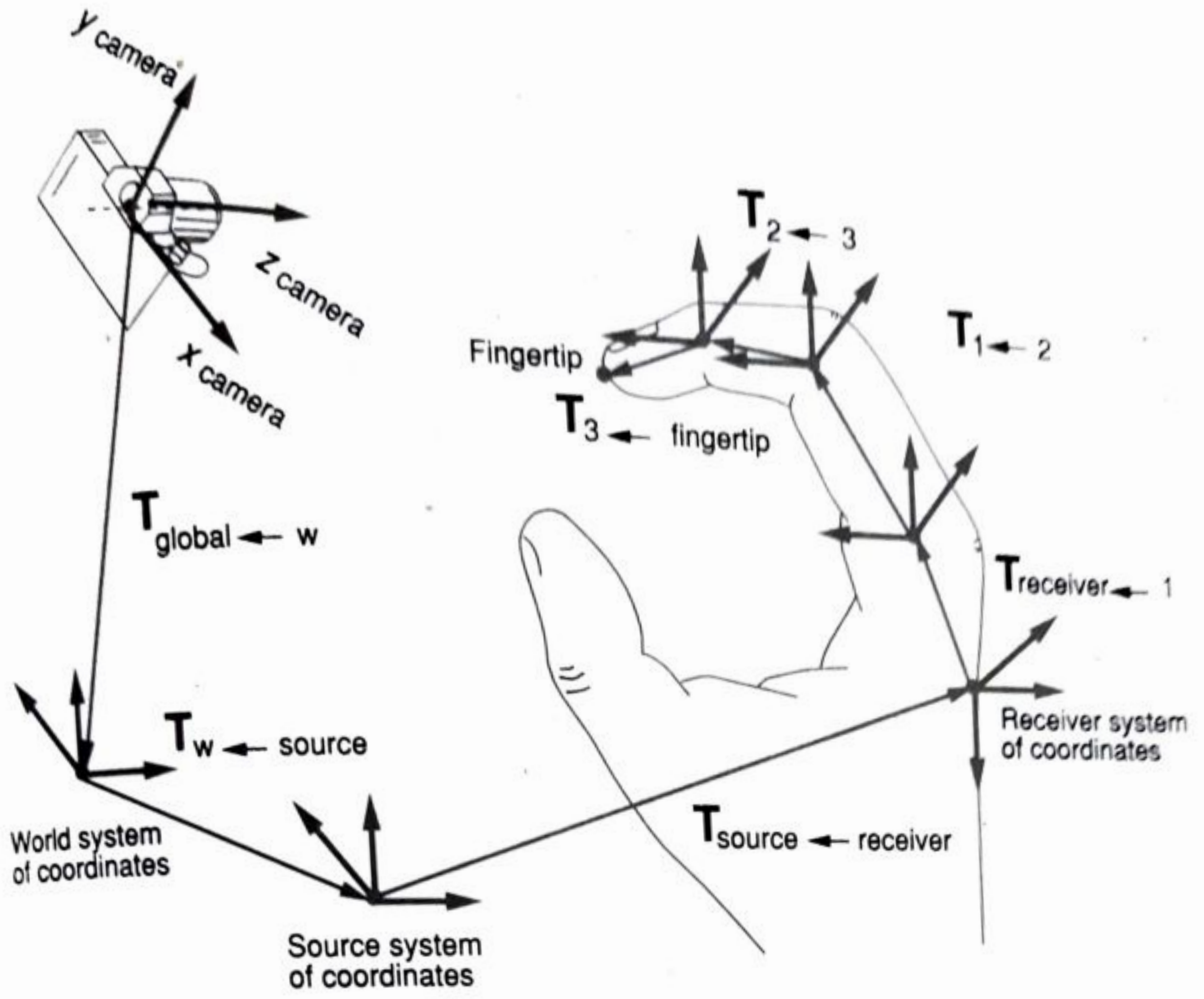


Fig. 5.18 The hierarchical structure of a virtual hand as seen by a virtual camera.

Viewing the Three-Dimensional World

Viewing the Three-Dimensional World

- The camera is oriented such that it is looking in the **positive z direction, y axis pointing upward and x axis to the right**
- The camera coordinates in **OpenGL is left-handed**
- The graphics pipeline is not concerned with the entirety of the virtual world, rather it processes only what the camera sees
- This represents a **volume of space called a frustrum**
- The **size of the projection is inversely proportional to the distance** from the object to the center of projection (distant objects appear small, closer ones large)

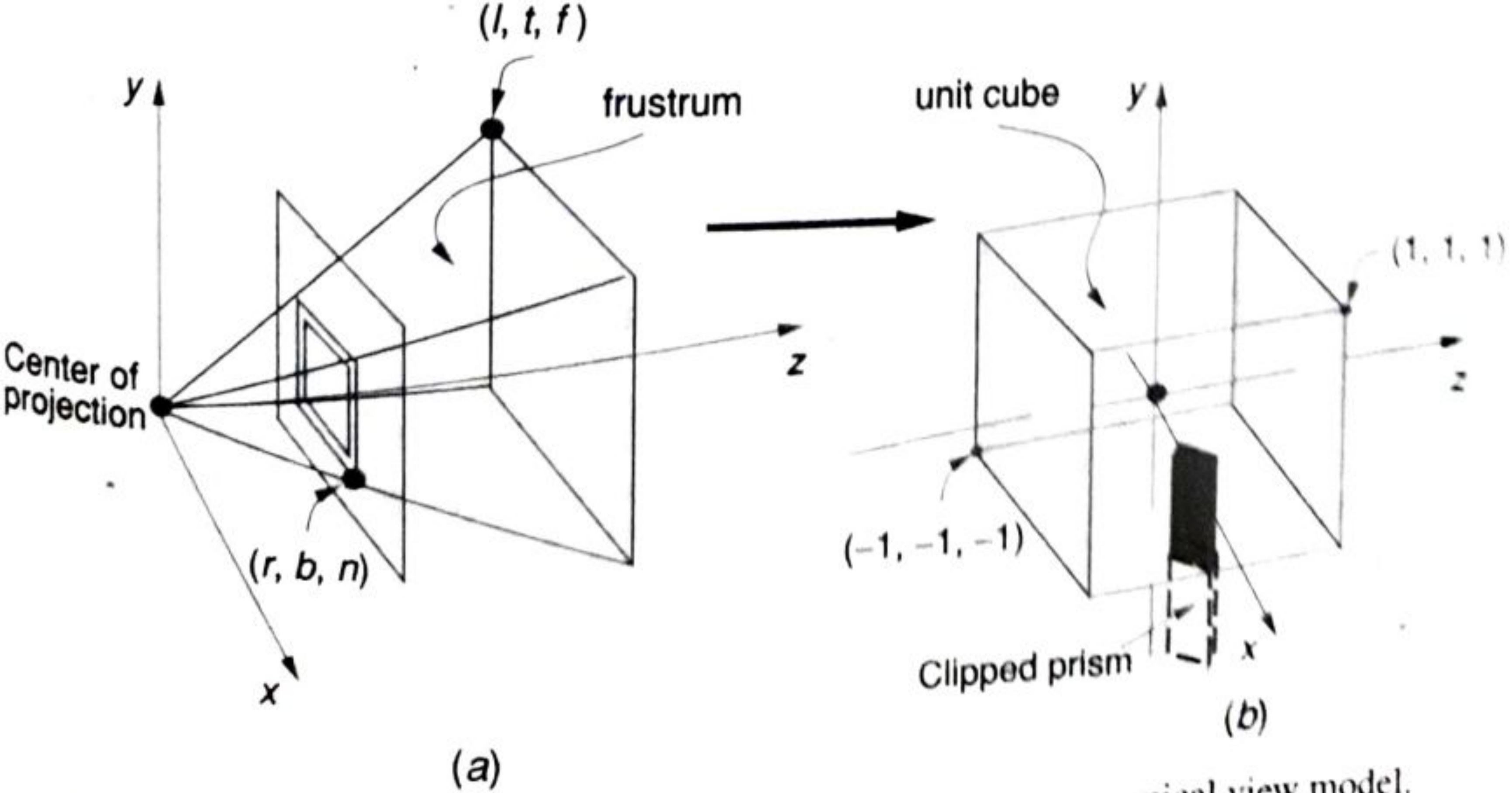


Fig. 5.19 Viewing the 3D World: (*a*) the frustrum; (*b*) the canonical view model.

canonical view volume

- This is a unit cube with corners at (-1, -1, -1) and (1, 1. 1).
- Once the 3D objects are projected into the canonical cube, their coordinates are normalized, which helps in the Z-buffering process
- Objects with higher z values are further away, and should not be rendered if occluded by objects with smaller z values
- The last stage of geometry processing is screen mapping. This stage translates and scales the (x, y) vertex coordinates of the objects inside the cube to the dimensions of a 2D display window

$$\mathbf{T}_{\text{projection}} = \begin{bmatrix} \frac{2n}{r-l} & 0 & \frac{r+l}{r-l} & 0 \\ 0 & \frac{2n}{t-b} & \frac{t+b}{t-b} & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Thank you